Dataset Datasheet

Course: DSPRO2 HS26 [review: template says DSPRO1/2 FS26]

Project Name:	Football Analyzer
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Date:	November 5, 2026
GitHub Repository:	`github.com/LittleBigFatFish/dspro2-sports-analyzer`

This sheet describes our main dataset, the Football Image Dataset. Five more datasets are compared in the table at the end.

Dataset Overview

- **Official Name:** Football Image Dataset (GitHub `skaczylo/Football-Image-Dataset`).
- **Owner / Contact:** GitHub user skaczylo; no other contact given.
- **Short Description:** 612 frames from football matches (512 train, 100 val) with 10,136 boxes for ball, player and referee in YOLO format. Frames come from TV broadcasts, highlight videos and one video game.
- **Version & Date:** No release tag. Commit `2642128` (2026-06-23), cloned 2026-10-20.
- **Reference:** `https://github.com/skaczylo/Football-Image-Dataset`
  BibTeX in the README, no paper.

Motivation and Intended Use

- **Primary Purpose:** A ready-to-use training set for Ultralytics YOLO (README). Funding is not stated.
- **Intended Use:** Detecting players, referees and the ball. We test pretrained YOLOv8n, YOLO12n and YOLO26n at 640 and 1280 px, later we fine-tune on it.
- **Out-of-Scope Use:** Identifying people; tracking or possession; claims about unseen matches; reporting video-game frames as real footage.

Composition and Coverage

- **Data Types:** Unstructured: JPEG images, one YOLO label file per image, `data.yaml` with the class names.
- **Population:** Professional matches 2024–25 from TV. Train: Bundesliga, Saudi league, J-League, Copa Libertadores, 2024 Community Shield. Val: three highlight clips and the video game eFootball PES 2021.
- **Dataset Size:** 612 images, 10,136 boxes (8,844 player, 792 referee, 500 ball). 548 images are 1920×1080, the rest smaller.
- **Known Gaps / Imbalances:** 43 of the 100 val images are from the video game. Balls are tiny: 74 of 83 val balls are under 8 px at 640 px input. No rain or snow, no amateur or youth games, no wide tactical camera.

Sources and Collection

- **Provenance:** Public third-party footage: TV broadcasts, YouTube highlights and one video game. The number of matches is unknown.
- **Collection Method:** Frames cut from videos. Sampling rule and tools are not documented.

- **Jurisdictions:** Broadcasts from Germany, Saudi Arabia, Japan, South America, England and two international matches. Where the data was labelled is not stated.

Labels and Targets

- **Target Definition:** Boxes for 0 ball, 1 player (goalkeepers included), 2 referee (linesmen included). For pretrained COCO models, player and referee count as *person*, ball as *sports ball*.
- **Labeling Process:** "Manually annotated" per the README; annotators and tool unknown.
- **Quality Control:** None documented upstream. Ours: the audit below and a spot-check of 35 images by Johan [review: error rate].

Quality Assessment (Data Audit)

- **Preprocessing Steps:** None; raw data is never edited. For inference, images are only resized to 640 or 1280 px.
- **Initial Audit Findings:** 0 errors, 31 warnings, among them 11 boxes under 5 px and one empty label on a frame full of players. 48 of the 100 val images are also in Soccana train, so a model trained on Soccana cannot be tested on this val split.
- **Quality Dimensions:** *Completeness:* every image has a label file. *Accuracy:* two label errors found, spot-check [review: result]. *Consistency:* class IDs match `data.yaml`; real and video-game frames are mixed. *Timeliness:* matches 2024–25; commit and file hashes recorded. Zero-shot at 1280 px on real val: person AP50 .95–.97, ball recall .35–.52 ($n = 46$ balls).

Legal, Ethical, and Governance

- **Sensitive Data:** Players, officials and fans are identifiable; there are no identity labels. The data stays on our machines and is not in git.
- **Compliance:** EU AI Act: minimal risk, since we do not identify people and it is coursework [review: confirm with supervisor]. GDPR / nFADP: the people shown are personal data; teaching use only, stored locally, not shared.
- **Consent & Licensing:** CC BY 4.0 per `LICENSE`. The frames come from broadcasts and a Konami game, so the licence covers the annotations at most. Consent is not mentioned.

Dataset Structure and Schema

- **File Formats:** JPEG, YOLO `.txt`, `data.yaml`.
- **Directory Layout:** `images/` and `labels/`, each with `train/` and `val/`.
- **Schema Summary:** Image and label share the base name; one line per box: `class x_center y_center width height`.
- **Units & Encodings:** Coordinates normalised to $[0, 1]$ by image size; UTF-8 text.

Other Datasets

	Soccana V1	**SN Tracking 2023**	**SN GSR 2025**	**SkillCorner**	**MultiSports (football)**
Data	11,673 images	79,500 frames, 106 clips	123,000 frames, 164 clips	20 matches, 1.24 M frames	534 clips, 305k frames
Labels	64,648 boxes	1,297,549 boxes	2,046,214 annotations	tracking, 94,361 events	8,297 action tubes
Classes	player, ball, referee	roles, team, jersey	as SN Tracking + pitch position	players, ball, events	15 actions (acting player only)
Licence	none	none; cite paper	GPL-3.0 tag	MIT	CC BY-NC 4.0
Our role	training only	main test; fine-tuning; tracking	test on valid; minimap	play analysis	person recall

Credits

- Football Image Dataset by skaczylo (CC BY 4.0).
- Soccana V1 by Adit-jain and the Roboflow Universe datasets on its card.
- SoccerNet-Tracking: A. Cioppa et al., CVPR Workshops 2022, arXiv:2204.06918.
- SoccerNet Game State Reconstruction: V. Somers et al., CVPR Workshops 2024, arXiv:2404.11335.
- SkillCorner Open Data by SkillCorner (with PySport), MIT.
- MultiSports by Y. Li et al., ICCV 2021, arXiv:2105.07404, CC BY-NC 4.0.
- YOLO weights and `ultralytics` by Ultralytics, AGPL-3.0.